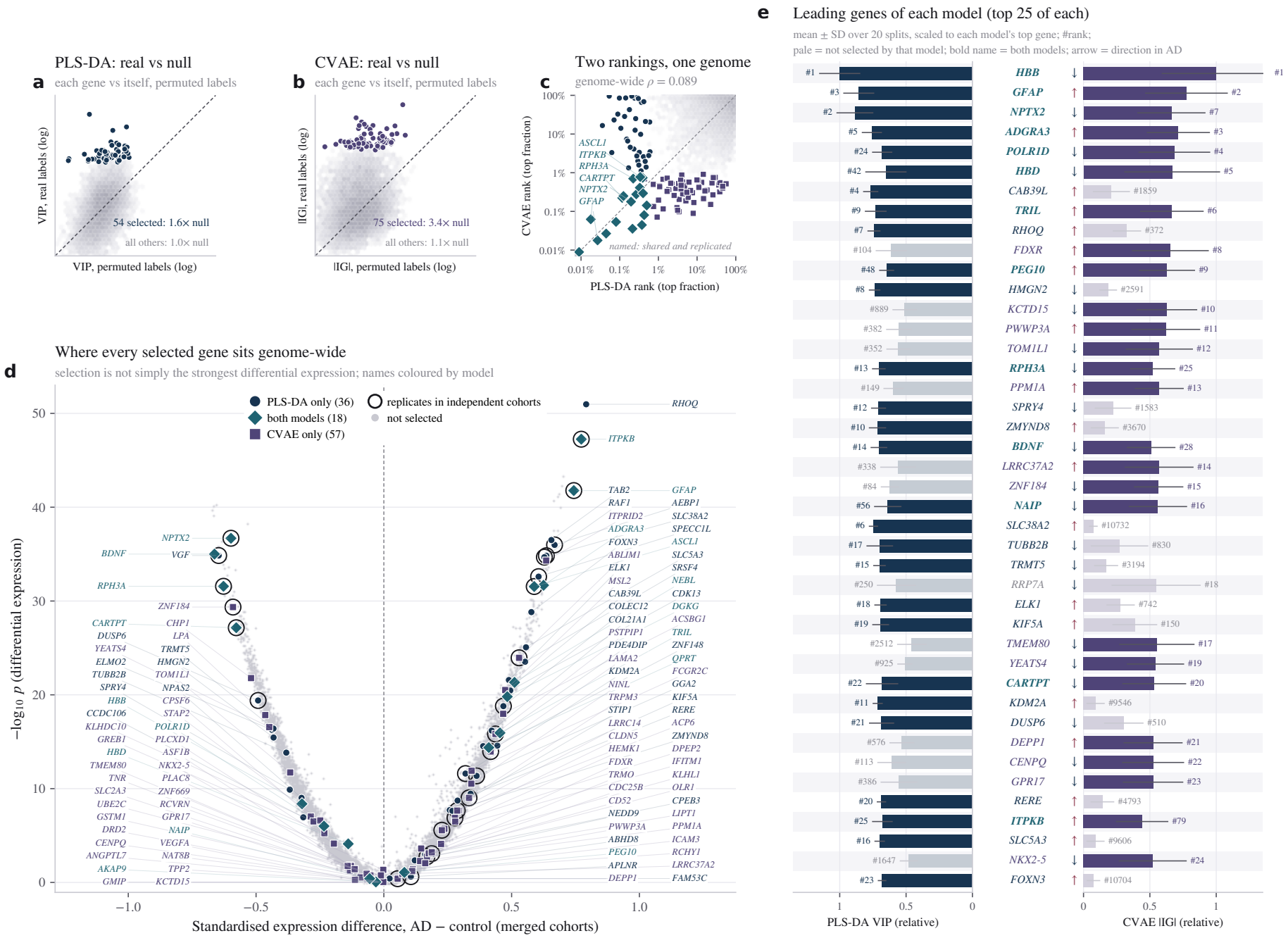

PLS-DA: real vs null
each gene vs itself, permuted labels
a
VIP, real labels (log)
VIP, permuted labels (log)
54 selected: 1.6× null
all others: 1.0× null
CVAE: real vs null
each gene vs itself, permuted labels
b
|IG|, real labels (log)
|IG|, permuted labels (log)
75 selected: 3.4× null
all others: 1.1× null
Two rankings, one genome
genome-wide ρ = 0.089
c
CVAE rank (top fraction)
PLS-DA rank (top fraction)
ASCL1
ITPKB
RPH3A
CARTPT
NPTX2
GFAP
named: shared and replicated
Where every selected gene sits genome-wide
selection is not simply the strongest differential expression; names coloured by model
d
PLS-DA only (36)
both models (18)
CVAE only (57)
replicates in independent cohorts
not selected
−log10 p (differential expression)
Standardised expression difference, AD − control (merged cohorts)
Leading genes of each model (top 25 of each)
mean ± SD over 20 splits, scaled to each model's top gene; #rank;
pale = not selected by that model; bold name = both models; arrow = direction in AD
e
PLS-DA VIP (relative)
CVAE |IG| (relative)